

FlowShield: AI-Powered Productivity & Focus Management Platform

Project Overview

FlowShield is a comprehensive productivity platform designed to help knowledge workers, developers, and students build better focus habits through structured work sessions, intelligent activity tracking, and actionable analytics. Unlike simple blockers or timers, FlowShield combines desktop-level activity monitoring with mobile and browser integration to provide a complete picture of productivity patterns, helping users understand *when* they work best, *what* distracts them, and *how* to optimize their focus time.

The MVP focuses on creating a frictionless experience for starting focus sessions, passively tracking work patterns, and delivering insights that drive behavioral change. By starting with individual users and proven productivity frameworks (Pomodoro technique), we can validate core value propositions before expanding to team features and advanced AI blocking in Phase 2.

Level

Medium to Hard

Type of Project

Full-Stack Application, Desktop Application, Browser Extension, Mobile Application, Productivity Tool, Analytics Platform

Skills Required

- **Backend:** Python/Node.js, FastAPI/Express, PostgreSQL/MongoDB
 - **Frontend:** React/Next.js, TailwindCSS
 - **Desktop App:** Electron or Tauri (Windows), native system hooks for activity tracking
 - **Browser Extensions:** Chrome Extension API, Firefox WebExtensions API
 - **Mobile:** React Native or Flutter (Android focus for MVP)
 - **Analytics:** Data visualization libraries (Chart.js, Recharts), time-series data handling
 - **DevOps:** Cloud hosting (AWS/GCP/Vercel), authentication (JWT/OAuth), API design
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Competitive Analysis

Market Landscape

Tool	Strengths	Weaknesses	FlowShield Differentiation
RescueTime	Mature desktop tracking, detailed reports, automatic categorization	No focus session structure, no mobile tracking integration, passive-only approach	FlowShield combines active session management with passive tracking; provides actionable coaching, not just reports
Toggl Track	Excellent time tracking, team features, project-based tracking	Manual tracking only, no distraction monitoring, no productivity insights	FlowShield automates tracking entirely; focuses on productivity quality, not just time logged
Forest	Beautiful gamification, phone usage deterrence, social features	No desktop tracking, no analytics, limited to phone-based focus	FlowShield covers all devices; provides real analytics instead of just gamification
Freedom	Powerful cross-platform blocking, scheduling, locked mode	Blocking-only approach, no productivity insights, no activity tracking	FlowShield provides understanding before restriction; insights-first approach
Focus@Will	Science-backed focus music, session timer	No activity tracking, no analytics, single-purpose tool	FlowShield is a complete productivity OS, not a single-feature tool
Centered	AI coaching, flow state detection, team features	Expensive (\$15/mo), limited free tier, desktop-only tracking	FlowShield offers more accessible pricing, mobile tracking, and a generous free tier

Competitive Positioning

FlowShield sits at the intersection of three categories that competitors typically address in isolation:

- 1. **Time Tracking** (Toggl, Clockify) → We add automatic tracking
- 2. **Distraction Blocking** (Freedom, Cold Turkey) → We add understanding before restriction
- 3. **Productivity Analytics** (RescueTime, Timing) → We add structured focus sessions and coaching

Our Unique Value: No existing tool combines structured focus sessions + passive multi-device tracking + AI-driven insights in a unified, privacy-first platform.

User Personas

Persona 1: Rafi — Freelance Web Developer

Attribute	Detail
Age	28
Location	Dhaka, Bangladesh
Work Style	Remote freelancer, works from home
Daily Schedule	Flexible hours, often 10 AM – 8 PM with gaps
Devices	Windows laptop, Android phone, Chrome browser
Pain Points	Constantly switches between client projects and social media; no structure to his day; underestimates time spent on distractions; struggles to bill accurately because he doesn't know his actual productive hours
Goals	Track real productive hours for better invoicing; build consistent deep work habits; reduce phone pickups during coding sessions
FlowShield Usage	Desktop app for tracking coding sessions; mobile app to monitor phone usage; weekly analytics to identify his peak coding hours and adjust schedule accordingly

Persona 2: Sneha — Computer Science Student

Attribute	Detail
Age	21
Location	Bangalore, India
Work Style	Full-time student, studies at home and library
Daily Schedule	Classes 9 AM – 2 PM, study sessions 4 PM – 10 PM
Devices	Windows laptop, Android phone
Pain Points	Can't focus during study sessions; YouTube and Instagram consume study time; feels guilty but doesn't know how bad the problem actually is; exam prep is always last-minute

Attribute	Detail
Goals	Build consistent study habits; see clear evidence of improvement; use streak mechanics for motivation; prepare for exams without cramming
FlowShield Usage	Pomodoro sessions for studying; distraction tracking to see actual time lost; goal system for weekly study hour targets; streak counter for daily motivation

Persona 3: David — Product Manager at a SaaS Startup

Attribute	Detail
Age	34
Location	Austin, Texas, USA
Work Style	Hybrid office/remote, meeting-heavy days
Daily Schedule	8 AM – 5 PM with meetings scattered throughout
Devices	MacBook (personal Windows at home), iPhone, Chrome browser
Pain Points	Days filled with shallow work (Slack, email, meetings); struggles to find blocks for deep thinking; context-switching destroys productivity; wants data to advocate for "no meeting" blocks with his team
Goals	Identify and protect deep work windows; quantify the cost of context switching; eventually bring FlowShield to his team (Phase 2 customer)
FlowShield Usage	Browser extension for tracking work patterns; desktop tracking to measure context switches; weekly reports to identify best focus windows; eventually champions team tier adoption

Persona 4: Maria — Content Creator & Writer

Attribute	Detail
Age	30
Location	Lisbon, Portugal
Work Style	Solo creator, writes articles and courses from home
Daily Schedule	Creative mornings, admin afternoons

Attribute	Detail
Devices	Windows desktop, Android tablet, Chrome browser
Pain Points	Creative blocks feel like wasted time; hard to distinguish "thinking time" from procrastination; research rabbit holes on the web; no way to measure creative productivity
Goals	Track creative vs. admin work separately; build a morning writing habit; understand her creative energy cycles; reduce research-to-writing ratio
FlowShield Usage	Custom session types (creative, admin, research); activity categorization to separate productive research from aimless browsing; weekly patterns to optimize her creative schedule

What is the App?

Core Concept

FlowShield is a productivity companion that helps users enter and maintain flow states through three integrated components:

1. **Focus Session Manager:** Structured work blocks with intelligent break scheduling
2. **Activity Monitor:** Passive tracking of computer and mobile usage during work sessions
3. **Productivity Intelligence:** Analytics engine that transforms raw activity data into actionable insights

The Problem We Solve

- **Awareness Gap:** Most people don't realize how fragmented their attention is until they track it
- **Planning Overhead:** Deciding when to work and for how long creates friction
- **Insight Deficiency:** Generic productivity advice doesn't account for individual work patterns
- **Multi-Device Reality:** Work happens across desktop, browser, and mobile — tracking needs to follow

Our Solution

FlowShield creates a "productivity operating system" that:

- Makes starting focused work effortless (one-click session start)
 - Automatically captures work patterns without manual logging
 - Delivers personalized insights about peak productivity times, common distractions, and focus trends
 - Provides gentle accountability through progress tracking and streak mechanics
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How Do I Use the App?

User Journey Flow

Initial Setup (5 minutes)

1. **Web Signup** → User creates account at app.FlowShield.com
2. **Onboarding Quiz** → Quick questions about work style:
 - Typical work hours (morning person vs. night owl)
 - Preferred focus duration (25/45/60/90 min blocks)
 - Primary distraction sources (social media, messaging, news, etc.)
 - Work environment (home/office/hybrid)
3. **Extension Install** → One-click install for Chrome/Firefox
4. **Desktop App Download** → Windows app for activity tracking
5. **Mobile App (Optional)** → Android app for phone activity tracking

First-Week Guided Experience (NEW)

To improve activation and reduce confusion, new users receive a structured onboarding during their first 7 days:

Day	Guided Action	Goal
Day 1	Complete first 25-min focus session with in-app tutorial overlay explaining timer, controls, and what's being tracked	Demystify the product
Day 2	Review "Your First Session" report with annotated explanations of productivity score, distraction count, and activity timeline	Teach users to read analytics
Day 3	Set a daily focus goal with contextual tooltip: "Most users like you start with 2 hours/day"	Introduce goal system
Day 4	Complete 2 sessions; receive first insight card: "Here's what we noticed about your focus pattern"	Demonstrate insight value
Day 5	Try a different session duration; tooltip explains why experimenting helps find optimal length	Encourage exploration
Day 6	View first weekly comparison (partial); guided walkthrough of analytics dashboard sections	Build dashboard habit
Day 7	Celebrate first week with summary email; prompt to invite a friend or set weekly goals	Retention hook

Each step uses contextual tooltips and progressive disclosure — users never see all features at once.

Daily Usage Pattern

Morning Ritual (2 minutes)

8:00 AM - Open FlowShield dashboard

- See yesterday's productivity score + insights
- Review today's planned focus blocks
- Click "Start Morning Focus" (defaults to user's preferred duration)

During Focus Session (25-90 minutes)

Desktop app runs in background:

- Tracks active window titles (anonymized)
- Monitors keyboard/mouse activity levels
- Detects context switches between applications

Browser extension:

- Logs tab navigation patterns
- Identifies productive vs. distracting domains
- Shows persistent timer in browser toolbar

Mobile app (if installed):

- Tracks phone unlocks during session
- Logs app usage time
- Sends gentle reminder if phone used excessively

Break Time (5-15 minutes)

Session ends → Desktop notification + browser alert

Break reminder with suggested activities:

- ☕ Get coffee/water
- 🚶 Short walk
- 👁 Eye rest (look 20 feet away)
- 🧘 Breathing exercise

User can:

- Start break timer
- Skip break and continue
- End session early

End of Day (3 minutes)

5:00 PM - Open dashboard to see:

- Total focus time achieved
- Productivity score (0-100)
- Top distractions of the day
- Focus time comparison vs. weekly average

Quick journal prompt:

"What helped you focus today?" (optional)

Weekly Review (10 minutes)

Sunday evening:

- Receive email with weekly summary
- Dashboard shows:
 - * Total focus hours
 - * Best focus day
 - * Peak productivity time blocks
 - * Distraction heatmap
 - * Goal progress (if set)
 - * Suggestions for next week

Platform-Specific Flows

Web Dashboard

- Primary interface for reviewing analytics
- Setting up focus goals and preferences
- Starting web-based focus sessions (for users without desktop app)
- Viewing historical data and trends

Desktop App (Windows)

- System tray icon for quick session start
- Minimal UI — primarily background process
- Local data collection with encrypted sync to cloud
- Native notifications for session transitions

Browser Extension

- Toolbar timer showing current session status

- Right-click context menu to start/pause session
- Badge showing focus streak
- Quick access to today's stats

Mobile App (Android)

- Widget for home screen (current session timer)
 - Push notifications for break reminders
 - Simplified session start (optimized for on-the-go)
 - Basic daily stats view
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What Are the Patterns Behind the App?

Behavioral Design Patterns

1. Friction Reduction

- **Pattern:** Make starting easier than procrastinating
- **Implementation:**
 - One-click session start from any platform
 - Pre-filled session durations based on history
 - Auto-start if user opens certain apps (optional)

2. Passive Data Collection

- **Pattern:** Track activity without manual logging
- **Implementation:**
 - Background monitoring requires zero user input
 - Activity is anonymized and categorized automatically
 - Users review insights, not raw logs

3. Habit Loop Design (Cue → Routine → Reward)

- **Cue:** Morning dashboard visit, scheduled notifications
- **Routine:** Focus session with break structure
- **Reward:** Productivity score, streak counter, progress visualization

4. Progressive Disclosure

- **Pattern:** Show simple metrics first, deeper insights over time
- **Implementation:**
 - Day 1: Basic timer and "time focused" metric
 - Week 1: Daily comparison and distraction categories
 - Month 1: Peak performance times and pattern recognition
 - Month 3: Personalized productivity recommendations

5. Social Proof (Light Touch)

- **Pattern:** Subtle comparison without unhealthy competition
- **Implementation:**
 - "You focused more than 67% of users today" (opt-in)
 - Anonymous leaderboard position
 - Community average stats

Technical Architecture Patterns

1. Event-Driven Activity Tracking

Desktop App (Local) → Activity Events → Local Buffer →
Cloud API (Batched) → Processing Queue → Analytics Engine → Dashboard

2. Progressive Web App + Native Sync

- Web app works offline with local storage
- Native apps sync when online
- Conflict resolution prioritizes most recent data

3. Privacy-First Data Model

- Raw activity data stored locally
- Only aggregated metrics sent to cloud
- User can view/export/delete all data
- Window titles are hashed with user-specific salt

4. Modular Components

Core Session Engine (shared logic)

- └─ Web Dashboard
- └─ Desktop App
- └─ Browser Extension
- └─ Mobile App

Each platform wraps the core engine with platform-specific UI and data collection.

How Do I Excel at the App?

For Users: Mastery Progression

Level 1: Beginner (Week 1-2)

- **Goal:** Establish baseline and build session habit
- **Success Metrics:**
 - Complete 3 focus sessions per day
 - Start sessions consistently (daily streak)
- **Key Insight:** Discover your actual focus capacity (many people overestimate)

Level 2: Practitioner (Week 3-8)

- **Goal:** Optimize session timing and duration
- **Success Metrics:**
 - Identify peak productivity windows
 - Reduce average distractions per session by 30%
 - Maintain 5-day streaks
- **Key Insight:** Learn your distraction triggers and pre-emptively manage them

Level 3: Expert (Month 3+)

- **Goal:** Achieve flow state consistently
- **Success Metrics:**
 - 4+ hours deep focus per day
 - 85+ productivity score regularly
 - Predictable energy/focus patterns
- **Key Insight:** You've internalized focus as a skill, not just a timer

Power User Features to Unlock

Advanced Analytics

- Correlation analysis: "You focus 40% better after morning exercise"
- Context detection: "Code reviews take 25% longer after lunch"
- Predictive scheduling: "Start important work Tuesday mornings"

Custom Session Types

- Deep Work (90 min, no breaks)
- Creative Work (50 min, optional ambient sound)
- Admin Tasks (25 min, short breaks)
- Learning Mode (45 min, review breaks)

Integration Workflows

- Calendar sync: Auto-start sessions for "Focus Time" blocks
- Task manager integration: Link sessions to specific projects
- API access: Export data to personal productivity systems

For Developers: Technical Excellence

Data Collection Best Practices

- Respect battery life: Sampling rate adapts to device state
- Privacy by design: Implement local filtering before cloud sync
- Efficient querying: Use time-series database for analytics
- Graceful degradation: App works even if one component fails

Analytics Pipeline Optimization

- Real-time dashboard updates via WebSockets
- Batch processing for daily/weekly reports
- Caching layer for frequently accessed metrics
- ML model training on aggregated user data (opt-in)

Extension Performance

- Minimize memory footprint (< 50MB)

- Lazy load analytics visualizations
 - Use service workers for background tasks
 - Optimize API calls with request batching
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Key Features & Milestones

Milestone 1: Core Focus Session System (Weeks 1-4)

1.1 Web Dashboard Foundation

- User authentication (email/password, Google OAuth)
- Basic profile setup and preferences
- Session control panel:
 - Start/pause/stop session buttons
 - Duration selector (25/45/60/90 min presets + custom)
 - Session type selector (work/study/creative)
- Real-time timer display
- Basic stats: Total focus time today, current streak

Technical Requirements:

- REST API with JWT authentication
- PostgreSQL database schema:
 - Users table
 - Sessions table (start_time, end_time, duration, type, productivity_score)
 - Settings table (user preferences)
- React frontend with responsive design
- Deployment on Vercel/Netlify + database on Supabase/Railway

1.2 Desktop Activity Tracker (Windows)

- System tray application
- Background process for activity monitoring:
 - Active window detection (using Win32 API)
 - Keyboard/mouse activity levels (events per minute)
 - Application usage tracking

- Local data storage with encryption
- Sync to cloud API (batched every 5 minutes)
- Session controls from system tray

Technical Requirements:

- Electron or Tauri framework
- Windows API integration (node-ffi or native modules)
- SQLite local database
- Encrypted sync protocol
- Auto-start on system boot

1.3 Browser Extension (Chrome/Firefox)

- Toolbar icon with timer display
- Active tab tracking during sessions
- Domain categorization (productive vs. distracting)
- Context menu integration ("Start Focus Session")
- Badge notifications for session events

Technical Requirements:

- Manifest V3 (Chrome) and V2 compatibility (Firefox)
 - Background service worker
 - Content scripts for tab monitoring
 - Local storage with cloud sync
 - Cross-browser compatibility layer
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Milestone 2: Mobile App & Break System (Weeks 5-8)

2.1 Android App

- Session timer with large, readable display
- Quick start widget for home screen
- Phone usage tracking during sessions:
 - Screen unlock count
 - App usage time (using UsageStatsManager)

- Notification interruptions
- Push notifications for break reminders
- Minimal stats view (today's focus time)

Technical Requirements:

- React Native or Flutter
- Android Usage Stats API
- Background service for tracking
- Firebase Cloud Messaging for notifications
- Local SQLite with cloud sync

2.2 Intelligent Break System

- Break timer with countdown
- Suggested break activities based on session duration:
 - 25 min work → 5 min break
 - 45 min work → 10 min break
 - 90 min work → 15 min break
- Break enforcement options:
 - Gentle reminder (notification only)
 - Moderate (desktop overlay)
 - Strict (block input — optional)
- Break activity suggestions with randomization

Technical Requirements:

- Timer logic in core session engine
 - Native notifications across all platforms
 - Optional desktop overlay (Electron)
 - User preference controls
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Milestone 3: Productivity Analytics Engine (Weeks 9-13)

3.1 Data Processing Pipeline

- Activity categorization engine:
 - Machine learning model to classify apps/websites as:
 - Deep Work (code editors, design tools)
 - Shallow Work (email, messaging, meetings)
 - Distractions (social media, entertainment)
 - Neutral (system apps, unknown)
- Productivity score calculation:

$$\text{Score} = (\text{Deep Work Time} \times 2 + \text{Shallow Work Time} - \text{Distraction Time} \times 2) / \text{Total Session Time} \times 100$$

- Time-series aggregation (hourly, daily, weekly)

Technical Requirements:

- Background job processor (Celery/Bull)
- Redis for caching
- Pre-trained ML model or rule-based classifier
- Time-series database (TimescaleDB or InfluxDB)

3.2 Analytics Dashboard

- **Daily View:**
 - Focus time vs. goal (progress bar)
 - Productivity score with trend indicator
 - Top 5 distractions (apps/websites with time spent)
 - Focus session timeline (visual blocks)
- **Weekly View:**
 - Daily focus time bar chart
 - Productivity score line graph
 - Best focus day + time
 - Distraction heatmap (day × hour grid)
- **Insights Panel:**
 - Automated observations:
 - "Your focus is 30% higher in mornings"
 - "Checking email reduces productivity for 15 min afterward"

- "Tuesdays are your most productive day"
- Goal progress tracking
- Streak counter with historical best

Technical Requirements:

- Recharts or Chart.js for visualizations
- Efficient API queries with pagination
- Real-time updates via WebSocket (optional)
- Export functionality (CSV/PDF)

3.3 Goal System

- Goal types:
 - Daily focus time target (hours/day)
 - Weekly focus time target
 - Productivity score target
 - Streak maintenance
 - Goal setting wizard
 - Progress tracking with visual indicators
 - Achievement notifications
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Milestone 4: Polish & Optimization (Weeks 14-16)

4.1 User Experience Refinements

- Onboarding flow optimization
- Keyboard shortcuts for power users
- Dark mode
- Customizable timer sounds
- Session history with search/filter
- Data export (all user data)

4.2 Performance & Reliability

- Frontend optimization (code splitting, lazy loading)

- API response time < 200ms
- Desktop app memory usage < 100MB
- Extension CPU usage < 5%
- Offline mode with sync conflict resolution
- Error tracking and crash reporting

4.3 Privacy & Security

- GDPR compliance features:
 - Data deletion on request
 - Privacy policy and terms
 - Cookie consent
 - Data anonymization for analytics
 - Penetration testing
 - Security audit of activity tracking
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App & Domain Classification System (NEW)

Overview

The classification system is the backbone of FlowShield's productivity scoring. Inaccurate classification directly undermines user trust, so this system needs careful design.

Three-Layer Classification Architecture

Layer 1: Default Classification Database (Seed Data)

A curated database of ~2,000 common applications and ~10,000 popular domains, pre-classified into four categories:

Category	Examples	Score Weight
Deep Work	VS Code, IntelliJ, Figma, Photoshop, Google Docs (active editing), terminal emulators	×2.0 positive
Shallow Work	Gmail, Slack, Microsoft Teams, Outlook, Zoom, Google Calendar, Notion (browsing)	×1.0 positive
Distraction	Facebook, Instagram, Twitter/X, TikTok, Reddit (non-work), YouTube (non-tutorial), Netflix, gaming apps	×2.0 negative
Neutral	System settings, file explorer, calculator, weather apps, OS processes	×0 (excluded)

Seed data sources:

- Manually curated list of top 500 apps and top 5,000 websites
- Open-source categorization datasets (e.g., domain classification lists)
- Community contributions (Phase 2)

Layer 2: User-Level Overrides

Users can override any classification for their personal context:

- YouTube → "Deep Work" (for a video editor or music producer)
- Reddit → "Shallow Work" (for a community manager)
- Twitter/X → "Deep Work" (for a social media marketer)

User overrides are stored per-user and always take priority over the default database.

UI for overrides:

- In the analytics dashboard, each app/domain entry has a "Reclassify" dropdown
- During onboarding, users can pre-classify their most-used tools
- Bulk import: Users can upload a list of domains with classifications

Layer 3: ML-Based Contextual Classification (Phase 2)

- Analyze usage patterns to suggest reclassifications: "You use Stack Overflow mostly during deep coding sessions — should we classify it as Deep Work?"
- Time-of-day context: YouTube at 2 PM during a study session vs. YouTube at 10 PM

- Session type context: The same app might be "Deep Work" in a "Creative" session but "Distraction" in a "Study" session

Classification Confidence & Transparency

Each classification includes a confidence indicator visible to the user:

Confidence Level	Source	User Action
High (green)	User-defined override	No action needed
Medium (yellow)	Default database match	Can override if incorrect
Low (gray)	Unknown app/domain, ML-inferred	Prompted to classify

When an unrecognized app or domain is detected:

1. Classify as "Neutral" by default (no impact on score)
2. After 3 occurrences, prompt user: "We noticed you use [app] during focus sessions. How should we classify it?"
3. Store the user's choice for future sessions

Classification API

```
GET /api/v1/classifications?domain=github.com
Response: { domain: "github.com", category: "deep_work", confidence: "high", source: "default" }

PUT /api/v1/classifications/override
Body: { domain: "reddit.com", category: "shallow_work", context: "work_subreddits" }

GET /api/v1/classifications/unresolved
Response: [ { domain: "notion.so", occurrences: 5, suggested: "deep_work" } ]
```

Notification Strategy & Deduplication (NEW)

The Problem

FlowShield runs on 4 platforms simultaneously. Without a unified notification policy, users receive duplicate alerts across desktop, browser extension, mobile, and email — leading to notification fatigue and eventual disabling of all notifications.

Notification Priority Hierarchy

Priority 1 (Active Platform): The platform the user is currently interacting with

Priority 2 (Secondary Platform): Other connected platforms

Priority 3 (Async): Email (never real-time, always batched)

Platform Detection & Suppression Rules

Scenario	Notification Channel	Suppressed Channels
Desktop app active + browser extension installed	Desktop app native notification	Browser extension, mobile
Only browser extension active	Browser extension badge + popup	Mobile (if desktop is idle)
No desktop/browser activity detected	Mobile push notification	Desktop, browser
User is on break	Mobile only (gentle)	Desktop, browser
Session end (all platforms)	Primary active platform	Others suppressed for 30 seconds, then sent if not acknowledged

Notification Types & Frequency Limits

Type	Max Frequency	Channel	User Can Disable?
Session start confirmation	Per session	Active platform only	No (critical)
Session end / break time	Per session	Active platform only	No (critical)
Break over reminder	1 per break	Active platform → mobile fallback	Yes
Phone usage warning	Max 2 per session	Mobile only	Yes
Daily summary	1 per day	Email or push (user choice)	Yes
Weekly report	1 per week	Email only	Yes
Streak at risk	1 per day	Push notification	Yes
Goal achieved	Per achievement	Active platform	Yes
Insight discovered	Max 1 per day	Dashboard badge only	Yes

OS-Level Integration

- **Windows Focus Assist:** Detect when Windows Focus Assist / Do Not Disturb is enabled; queue notifications instead of suppressing them permanently
- **Android DND Mode:** Respect Do Not Disturb settings; use notification channels so users can selectively enable FlowShield during DND
- **Browser permissions:** Gracefully handle cases where notification permission is denied; fall back to in-page alerts

User Notification Preferences

Settings page includes:

- Master notification toggle per platform
- Granular controls per notification type
- "Quiet hours" configuration (e.g., no notifications after 10 PM)
- Notification sound selection (or silent)
- Vibration pattern for mobile (or off)

Offline Resilience & Sync Architecture (NEW)

Problem Statement

With 4 client platforms generating data independently, offline scenarios are inevitable — poor internet at a café, airplane mode during travel, or simply intermittent connectivity. The sync system must handle these gracefully without data loss.

Offline Capabilities by Platform

Platform	Offline Behavior	Local Storage
Web Dashboard	Read-only cached analytics; cannot start sessions (unless PWA)	IndexedDB (last 7 days of stats)
Desktop App	Full functionality; sessions and tracking continue normally	SQLite (unlimited local history)
Browser Extension	Sessions continue; tab tracking works; badge updates locally	Extension local storage (current session + 24hr buffer)

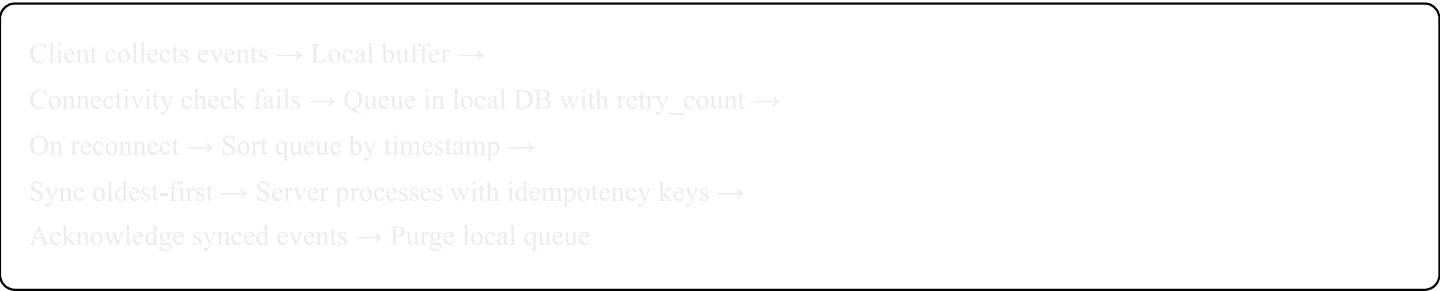
Platform	Offline Behavior	Local Storage
Mobile App	Sessions continue; phone tracking works; stats cached locally	SQLite (unlimited local history)

Sync Protocol

Batch Sync (Normal Operation)



Offline Queue (No Connectivity)



Conflict Resolution Strategy

Principle: Last-write-wins for settings/preferences; merge for activity data; server-authoritative for session state.

Data Type	Conflict Scenario	Resolution
Activity logs	Duplicate events from multiple clients	Deduplicate by <u>event_id</u> (UUID generated client-side); merge unique events
Session state	Session started on desktop, paused on browser	Server maintains canonical session state; latest action wins with timestamp comparison
User settings	Preferences changed on two devices offline	Last-write-wins with <u>updated_at</u> timestamp; user notified of conflict
Goals	Goal progress updated offline on two devices	Sum of increments (both contributions count); server reconciles totals
Classifications	User reclassifies same domain differently on two devices	Last-write-wins; display notification: "Your classification for X was updated on another device"

Data Integrity Guarantees

- Every event has a client-generated UUID (idempotency key)
- Server rejects duplicate UUIDs, returns success (idempotent)
- Sync acknowledgments are atomic — client only purges events after server confirms
- Maximum offline buffer: 72 hours of data (configurable); beyond this, oldest events are aggregated into hourly summaries to save space
- Health check endpoint: `GET /api/v1/sync/status` returns last sync time and pending event count

Data Retention & Storage Policy (NEW)

Retention Tiers

Data Type	Free Tier Retention	Pro Tier Retention	Storage Format
Raw activity logs	7 days	90 days	Time-series DB (TimescaleDB)
Session records	30 days	Unlimited	PostgreSQL
Daily aggregated stats	30 days	Unlimited	PostgreSQL
Weekly summaries	90 days	Unlimited	PostgreSQL
Monthly summaries	1 year	Unlimited	PostgreSQL
User settings/preferences	Unlimited	Unlimited	PostgreSQL
Local raw data (desktop/mobile)	Unlimited (user's device)	Unlimited (user's device)	SQLite

Automated Cleanup Jobs

- **Nightly job (2 AM UTC):** Aggregate raw activity logs older than retention period into daily summaries, then purge raw logs
- **Weekly job (Sunday 3 AM UTC):** Generate weekly summaries from daily stats
- **Monthly job (1st of month):** Generate monthly summaries; purge expired daily stats for free tier users
- **On-demand:** User-triggered data export before deletion

Storage Cost Projections

Scale	Monthly Raw Logs	After Aggregation	Estimated Monthly Cost
1,000 users	~50 GB	~5 GB	~\$15 (managed DB)
10,000 users	~500 GB	~50 GB	~\$120 (managed DB)
100,000 users	~5 TB	~500 GB	~\$800 (sharded DB)

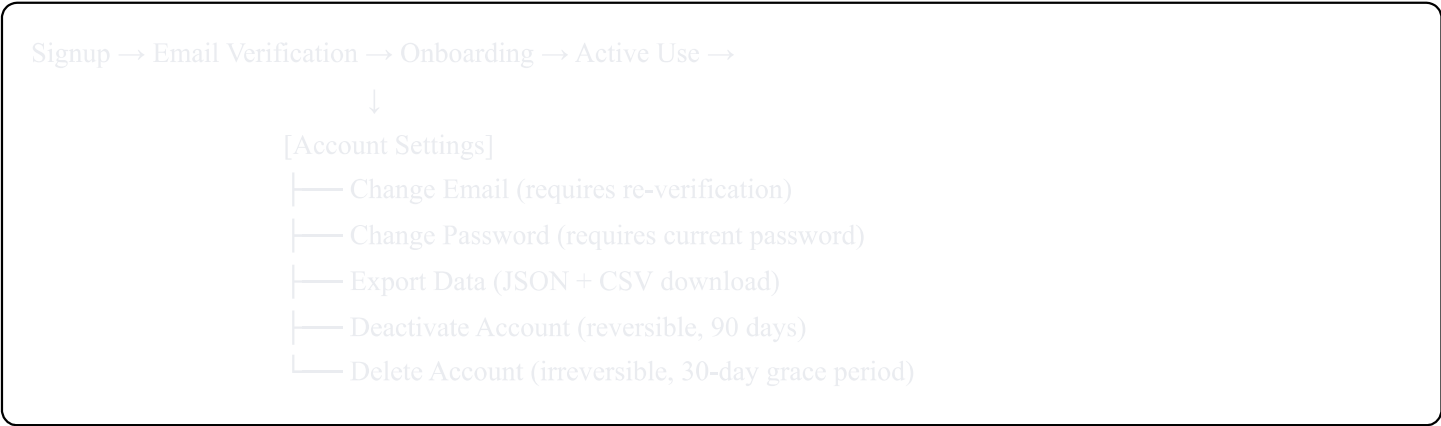
Assumptions: Average 4 sessions/day, 50 activity events per session, 200 bytes per event.

User Data Controls

- **Export:** One-click export of all user data in JSON + CSV format (GDPR Article 20)
- **Selective deletion:** Delete specific sessions, date ranges, or all data
- **Account deletion:** Full purge within 30 days, immediate anonymization of aggregated contributions
- **Local data:** Users can clear local SQLite databases independently from cloud data

Account Management & User Flows (NEW)

Account Lifecycle



Account Deletion Flow

1. User clicks "Delete Account" in settings
2. Confirmation modal: "This will permanently delete all your data. This cannot be undone."
3. User must type "DELETE" and enter password to confirm
4. Immediate actions:
 - Account marked as `pending_deletion`

- All active sessions terminated
- Desktop/mobile apps sign out on next sync
- Login disabled immediately

5. 30-day grace period:

- User receives email: "Your account is scheduled for deletion on [date]. Click here to cancel."
- During this period, user can log in to cancel deletion only
- No new sessions or data collection

6. After 30 days:

- All personal data permanently purged
- Aggregated anonymous statistics retained (GDPR-compliant)
- Email added to suppression list (no marketing)

Account Deactivation (Pause) Flow

For users who want to take a break without losing data:

1. Account marked as deactivated
2. All clients stop tracking; sessions cannot be started
3. Data is preserved; analytics remain viewable (read-only)
4. After 90 days of inactivity, user receives prompt to reactivate or delete
5. Reactivation is instant — login restores full functionality

Password Reset Flow

1. User clicks "Forgot Password" on login screen
2. Email sent with time-limited reset link (expires in 1 hour)
3. Link opens password reset form (new password + confirmation)
4. On success: All existing sessions invalidated; user must log in again on all devices
5. Rate limiting: Max 3 reset requests per hour per email

API Design & Developer Specification (NEW)

API Overview

RESTful API with JSON payloads. All endpoints require authentication via JWT Bearer token unless otherwise noted.

Base URL: `https://api.flowshield.com/v1`

Versioning: URL-based versioning (`/v1/`, `/v2/`). Deprecated versions supported for 12 months with deprecation headers.

Authentication

POST `/auth/register` — Create account (email + password)
POST `/auth/login` — Get access + refresh token
POST `/auth/refresh` — Refresh access token
POST `/auth/forgot-password` — Request password reset
POST `/auth/reset-password` — Complete password reset
POST `/auth/google` — OAuth2 Google sign-in
DELETE `/auth/logout` — Invalidate tokens

Core Endpoints

Sessions

POST `/sessions` — Start a new focus session
GET `/sessions` — List sessions (paginated, filterable by date/type)
GET `/sessions/:id` — Get session details
PATCH `/sessions/:id` — Update session (pause, resume, end)
DELETE `/sessions/:id` — Delete a session record
GET `/sessions/active` — Get currently active session (if any)

Activity

POST `/activity/batch` — Upload batch of activity events (from clients)
GET `/activity/:sessionId` — Get activity log for a session
GET `/activity/summary` — Get aggregated activity summary (daily/weekly/monthly)

Analytics

GET `/analytics/daily` — Daily stats (focus time, score, distractions)
GET `/analytics/weekly` — Weekly summary with trends
GET `/analytics/insights` — AI-generated insight cards
GET `/analytics/heatmap` — Focus heatmap data (day × hour)
GET `/analytics/trends` — Long-term trend data (weeks/months)

Goals

POST /goals — Create a new goal

GET /goals — List active goals

PATCH /goals/:id — Update goal target/status

DELETE /goals/:id — Delete a goal

GET /goals/:id/progress — Get goal progress history

Classifications

GET /classifications — Get classification database for user

GET /classifications/search — Search by domain or app name

PUT /classifications/override — Set user-level override

DELETE /classifications/override/:id — Remove override

GET /classifications/unresolved — Get unclassified apps/domains

User & Settings

GET /user/profile — Get user profile

PATCH /user/profile — Update profile

GET /user/settings — Get all settings

PATCH /user/settings — Update settings

POST /user/export — Request data export (async, returns download link)

DELETE /user/account — Request account deletion

Sync

POST /sync/batch — Client data sync (idempotent)

GET /sync/status — Get sync status and pending count

POST /sync/resolve-conflict — Manually resolve a flagged conflict

Rate Limiting

Endpoint Group	Rate Limit	Window
Auth endpoints	10 requests	15 minutes
Activity batch upload	60 requests	5 minutes
Analytics reads	120 requests	1 minute
Session CRUD	60 requests	1 minute
Classification overrides	30 requests	1 minute

Endpoint Group	Rate Limit	Window
Data export	3 requests	1 hour

Rate limit headers included in every response:

```
X-RateLimit-Limit: 120
X-RateLimit-Remaining: 117
X-RateLimit-Reset: 1708192800
```

Webhook Support (Phase 2)

```
POST /webhooks      — Register a webhook URL
GET  /webhooks      — List registered webhooks
DELETE /webhooks/:id — Remove a webhook
```

Events:

- session.started
- session.completed
- session.abandoned
- goal.achieved
- insight.generated
- streak.milestone (7, 30, 100 days)

Error Response Format

```
json
{
  "error": {
    "code": "SESSION_ALREADY_ACTIVE",
    "message": "Cannot start a new session while another is active.",
    "details": {
      "active_session_id": "uuid-here",
      "started_at": "2026-01-15T09:00:00Z"
    }
  }
}
```

Standard HTTP status codes: 200, 201, 400, 401, 403, 404, 409, 429, 500.

Security Model (NEW)

Desktop App Security

The desktop app has elevated system access for activity tracking. This creates a unique security surface that requires specific mitigations.

Code Signing & Distribution

- Windows app signed with EV code signing certificate (reduces SmartScreen warnings from ~80% to ~5%)
- Distributed via official website with SHA-256 checksum verification
- Auto-update mechanism with signed update packages
- No sideloading from third-party sources

Antivirus Compatibility

- Known issue: Electron/Tauri apps with system hooks trigger false positives in Windows Defender, Avast, Kaspersky, and others
- Mitigation strategy:
 - Submit to Microsoft, Avast, Kaspersky, and Malwarebytes for allowlisting before launch
 - Include clear documentation for users on how to add exceptions
 - Use well-known system APIs (SetWinEventHook) instead of low-level keyboard/mouse hooks where possible
 - In-app help page for common antivirus false positive resolution

Local Data Encryption

- SQLite database encrypted with AES-256
- Encryption key derived from user password + machine-specific salt (DPAPI on Windows)
- If user is not logged in, local data is inaccessible
- Data at rest is encrypted; data in memory is cleared on app close

Permission Model

- App requests only necessary permissions (accessibility API for window tracking, usage stats for app names)
- No keylogging — keyboard activity is measured as "events per minute" aggregate, not individual keystrokes
- No screen capture or screenshot capability

- Permissions explained clearly during installation with "Why we need this" tooltips

Network Security

- All API communication over TLS 1.3
- Certificate pinning for API endpoints
- No telemetry or analytics data sent to third parties
- Local-first: raw data never leaves the device; only aggregated metrics are synced

Browser Extension Security

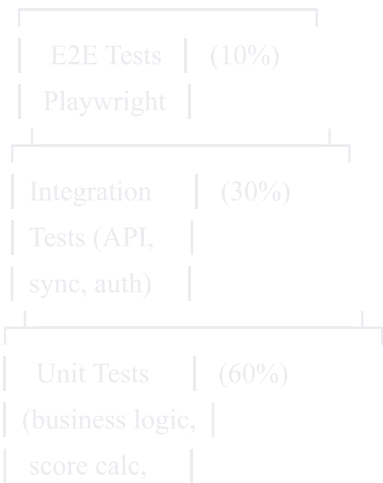
- Manifest V3 (Chrome) with minimal permissions: `activeTab`, `storage`, `alarms`, `notifications`
- No `<all_urls>` permission — tab URL tracking uses the `tabs` permission with user consent
- Content Security Policy prevents injection attacks
- Extension reviewed by Chrome Web Store and Firefox Add-ons teams

Mobile App Security

- Android app uses UsageStatsManager (requires explicit user permission grant in Settings)
- No root access required
- Local data encrypted with Android Keystore
- No background location tracking

Testing Strategy (NEW)

Testing Pyramid



Testing by Platform

Platform	Unit Tests	Integration Tests	E2E Tests	Manual QA
Backend API	Jest/Vitest for business logic, score calculation, classification engine	API endpoint tests with supertest; database integration tests	Full user flow tests via API	Load testing with k6
Web Dashboard	React component tests with Testing Library	API integration with MSW (Mock Service Worker)	Playwright browser tests (Chrome, Firefox, Edge)	Cross-browser visual QA
Desktop App	Core logic unit tests	Electron/Tauri integration tests; sync protocol tests	Manual testing on Windows 10, 11	Antivirus compatibility testing
Browser Extension	Extension logic unit tests	Chrome Extension API mocks	Manual testing on Chrome 120+, Firefox 120+	Extension store review compliance
Mobile App	Widget/screen unit tests	Android API integration tests	Detox / Maestro automated UI tests	Testing on Android 10, 11, 12, 13, 14

Platform Support Matrix

Platform	Minimum Version	Tested Versions
Windows	Windows 10 (build 1903+)	Windows 10, Windows 11
Chrome	Version 120+	Latest stable, latest - 1
Firefox	Version 120+	Latest stable, latest - 1
Android	Android 10 (API 29)	Android 10, 11, 12, 13, 14
Browsers (web)	Chrome, Firefox, Edge, Safari (latest)	Latest stable versions

CI/CD Pipeline

- All PRs require passing unit + integration tests
- E2E tests run on staging before every production deploy
- Desktop app builds tested on clean Windows VMs
- Mobile builds tested on BrowserStack / Firebase Test Lab

Performance Benchmarks (Automated)

Metric	Target	Test Method
API response time (p95)	< 200ms	k6 load test
Dashboard initial load	< 2s	Lighthouse CI
Desktop app memory usage	< 100MB	Automated memory profiling
Browser extension memory	< 50MB	Chrome DevTools protocol
Extension CPU usage	< 5%	Chrome task manager monitoring
Mobile app startup time	< 1.5s	Automated launch profiling
Sync batch processing	< 500ms for 100 events	API integration test

Product Analytics & Instrumentation (NEW)

Overview

Separate from user-facing analytics, FlowShield needs internal product analytics to understand feature adoption, identify friction points, and measure business metrics.

Analytics Tool

Primary: PostHog (self-hosted or cloud) — chosen for privacy-first approach and open-source option.

Fallback: Mixpanel or Amplitude if PostHog doesn't scale.

Key Events to Track

Onboarding Funnel

signup_started → signup_completed → onboarding_quiz_started →
onboarding_quiz_completed → extension_installed → desktop_app_installed →
mobile_app_installed → first_session_started → first_session_completed

Core Usage

session_started (with: duration_preset, session_type, platform)
session_completed (with: actual_duration, productivity_score, platform)
session_abandoned (with: time_elapsed, reason_if_provided)
break_started / break_skipped / break_extended
dashboard_viewed (with: section, time_spent)
insight_card_viewed / insight_card_dismissed
goal_created / goal_achieved / goal_abandoned
classification_overridden (with: from_category, to_category)

Retention & Engagement

daily_active (unique per day)
weekly_active (unique per week)
streak_length (on each session)
platforms_used (combination of active platforms)
notification_received / notification_clicked / notification_dismissed
data_exported / account_deleted

Key Dashboards (Internal)

1. **Activation Dashboard:** Signup → First Session conversion funnel
2. **Retention Dashboard:** D1, D7, D14, D30 retention curves
3. **Engagement Dashboard:** Sessions per user, average session duration, platform mix
4. **Feature Adoption:** Usage rates for goals, insights, custom sessions, classifications
5. **Churn Indicators:** Users approaching streak loss, declining session frequency, uninstalls

Accessibility (a11y) Requirements (NEW)

Target Compliance

WCAG 2.1 Level AA across all platforms.

Web Dashboard

- Full keyboard navigation for all interactive elements
- Focus indicators visible on all focusable elements
- ARIA labels on all icons, charts, and interactive components
- Color contrast ratio $\geq 4.5:1$ for normal text, $\geq 3:1$ for large text
- Charts include text alternatives (data tables accessible via toggle)
- Screen reader announcements for timer updates (every minute, not every second)
- Reduced motion support: Disable animations when `prefers-reduced-motion` is set
- Skip navigation links on dashboard

Desktop App

- System tray context menu accessible via keyboard
- Notifications use OS accessibility features
- High contrast mode support

Browser Extension

- Popup UI fully keyboard navigable
- Badge text readable by screen readers
- Context menu items have descriptive labels

Mobile App

- TalkBack (Android) support for all screens
 - Touch targets $\geq 48dp$
 - Content descriptions on all images and icons
 - Dynamic font size support (respect system font scaling)
-

Localization Strategy (NEW)

MVP (Phase 1)

- English only (US English as default)
- All user-facing strings externalized into i18n resource files from day one (no hardcoded strings)
- Date, time, and number formatting respects user's locale setting

- Timezone-aware throughout (stored as UTC, displayed in user's timezone)

Phase 2 Localization Targets

Priority	Language	Rationale
1	Bengali (বাংলা)	Target market — Bangladesh developer community
2	Hindi (हिन्दी)	Large South Asian user base
3	Spanish (Español)	Major global language
4	Portuguese (Português)	Growing tech community in Brazil
5	Arabic (العربية)	RTL support adds technical complexity but large addressable market

Technical Preparation

- Use `react-i18next` for web dashboard
 - Use platform-native `i18n` for mobile (Android string resources)
 - Browser extension uses `chrome.i18n` / `browser.i18n`
 - RTL layout support designed into CSS architecture from MVP (flexbox/grid that flips automatically)
 - Date formatting: Use `Intl.DateTimeFormat` everywhere — never hardcode date formats
-

Success Metrics

User Engagement

- **Activation:** 70% of signups complete first focus session within 24 hours
- **Retention:** 40% weekly active users (WAU) after 4 weeks
- **Habit Formation:** 25% of users maintain 7-day streak within first month
- **Session Completion:** 80% of started sessions reach completion

Product Performance

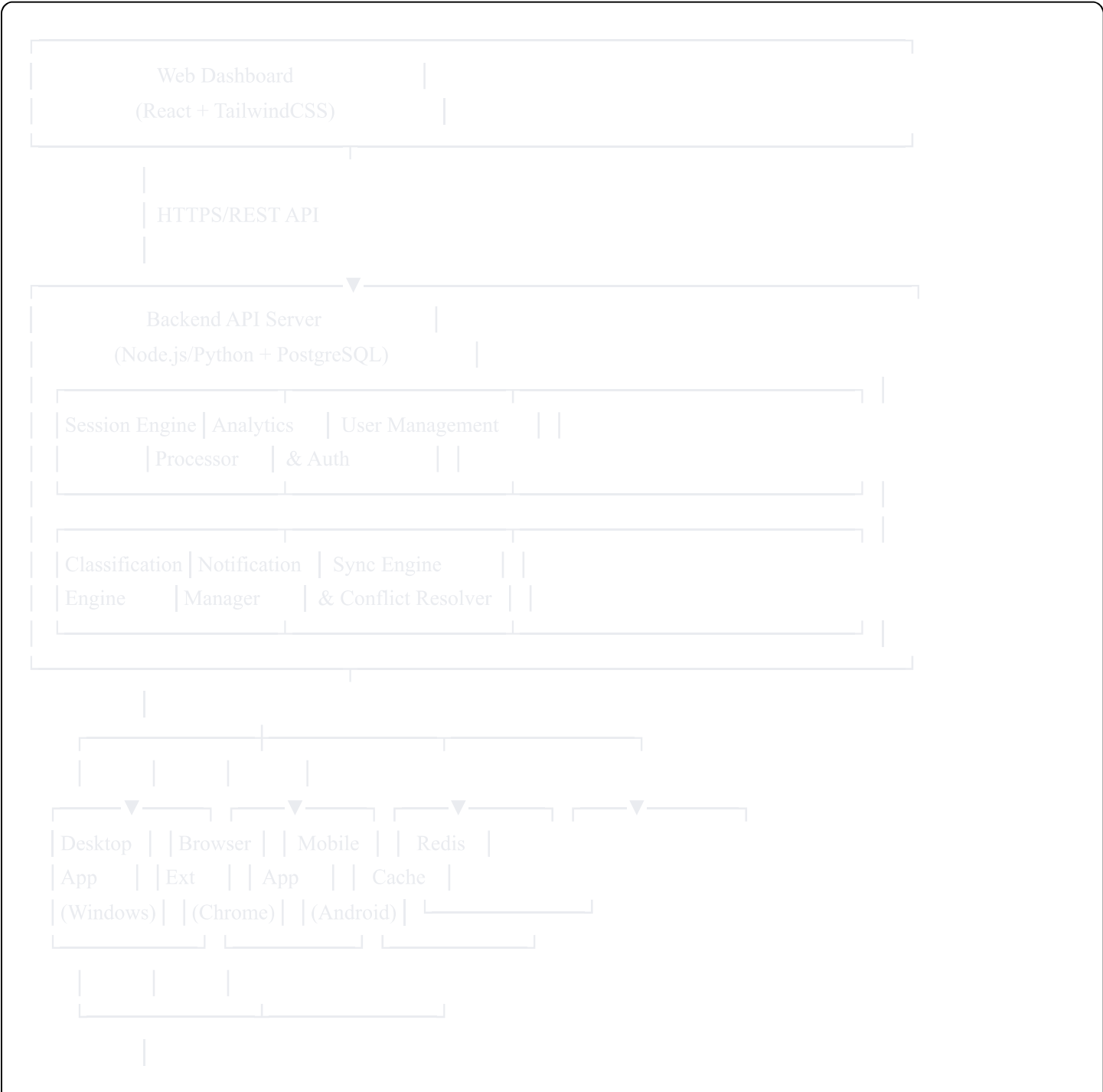
- **System Reliability:** 99.5% uptime for web services
- **Data Accuracy:** < 5% discrepancy in tracked vs. actual usage time
- **Load Times:** Dashboard loads in < 2 seconds
- **Extension Performance:** < 50MB memory usage

User Satisfaction

- **NPS Score:** Target 40+ within 3 months
- **Support Tickets:** < 5% of active users require support
- **App Store Rating:** 4.2+ stars on Chrome Web Store, Google Play

Technical Architecture

System Components



Data Flow

- 1. **Session Start:** User initiates session from any client
- 2. **Activity Tracking:** Local clients monitor activity and buffer data
- 3. **Sync to Cloud:** Batched uploads every 5 minutes (or immediately on session end)
- 4. **Processing:** Backend categorizes activity and calculates metrics
- 5. **Analytics Update:** Dashboard receives real-time or cached results
- 6. **Insights Generation:** Nightly batch job analyzes patterns and generates insights

Database Schema (Key Tables)

sql

-- Users

```
users {
  id: uuid PRIMARY KEY
  email: string UNIQUE
  hashed_password: string
  created_at: timestamp
  timezone: string
  locale: string DEFAULT 'en-US'
  preferences: jsonb
  status: enum (active, deactivated, pending_deletion)
  deletion_scheduled_at: timestamp NULL
}
```

-- Focus Sessions

```
sessions {
  id: uuid PRIMARY KEY
  user_id: uuid REFERENCES users
  start_time: timestamp
  end_time: timestamp
  planned_duration: integer (minutes)
  actual_duration: integer (minutes)
  session_type: enum (work, study, creative, admin, custom)
  productivity_score: integer (0-100)
  completed: boolean
  platform: enum (web, desktop, browser_ext, mobile)
}
```

-- Activity Logs (time-series data)

```
activity_logs {
  id: uuid PRIMARY KEY
  event_id: uuid UNIQUE (client-generated, for idempotent sync)
  session_id: uuid REFERENCES sessions
  timestamp: timestamp
  source: enum (desktop, browser, mobile)
  app_name: string (hashed)
  category: enum (deep_work, shallow_work, distraction, neutral)
  classification_confidence: enum (high, medium, low)
  duration_seconds: integer
  activity_level: integer (0-100)
  synced_at: timestamp
}
```

-- Classification Overrides

```
classification_overrides {
  id: uuid PRIMARY KEY
  user_id: uuid REFERENCES users
  domain_or_app: string
  category: enum (deep_work, shallow_work, distraction, neutral)
  context: string NULL (optional note)
  created_at: timestamp
  updated_at: timestamp
}

-- Goals
goals {
  id: uuid PRIMARY KEY
  user_id: uuid REFERENCES users
  goal_type: enum (daily_time, weekly_time, streak, productivity_score)
  target_value: integer
  current_value: integer
  start_date: date
  active: boolean
}

-- Daily Stats (aggregated)
daily_stats {
  id: uuid PRIMARY KEY
  user_id: uuid REFERENCES users
  date: date
  total_focus_minutes: integer
  avg_productivity_score: integer
  top_distractions: jsonb
  peak_focus_hour: integer
  sessions_completed: integer
  sessions_abandoned: integer
  total_context_switches: integer
  phone_unlocks: integer NULL
}

-- Notification Preferences
notification_preferences {
  id: uuid PRIMARY KEY
  user_id: uuid REFERENCES users
  channel: enum (desktop, browser, mobile, email)
  enabled: boolean DEFAULT true
  quiet_hours_start: time NULL
  quiet_hours_end: time NULL
}
```

```
notification_types: jsonb (per-type enable/disable)
}

-- Sync Status
sync_status {
  id: uuid PRIMARY KEY
  user_id: uuid REFERENCES users
  platform: enum (desktop, browser, mobile)
  last_sync_at: timestamp
  pending_events: integer
  last_conflict_at: timestamp NULL
}
```

Monetization Strategy (Post-MVP)

Freemium Model

Free Tier:

- Unlimited focus sessions
- 7 days of analytics history
- Basic productivity score
- 1 device sync

Pro Tier (\$8/month or \$72/year):

- Unlimited analytics history
- Advanced insights and correlations
- Unlimited device sync
- Custom session types
- Goal tracking with unlimited goals
- Priority support
- Data export

Teams Tier (\$15/user/month) (Phase 2)

- All Pro features
- Team dashboards
- Shared goals and challenges

- Admin controls
 - Team analytics
-

Phase 2 Features (Post-MVP)

Smart Website Blocking

- AI-powered distraction detection
- Context-aware blocking (e.g., allow YouTube for tutorial watching)
- Custom block lists with scheduling
- "Nuclear option" mode (hardcore blocking)

Cross-Platform Sync

- Seamless real-time sync across all devices
- Conflict resolution
- Offline mode with queue sync

Team Collaboration

- Shared focus goals
- Team productivity leaderboards
- Group challenges and accountability
- Team analytics dashboard

AI Productivity Coach

- Personalized recommendations based on patterns
 - Predictive scheduling ("Schedule deep work for Tuesday 9 AM")
 - Natural language insights
 - Proactive distraction warnings
-

Risks & Mitigation

Risk	Impact	Likelihood	Mitigation
Privacy concerns with activity tracking	High	High	Clear communication, local-first approach, anonymization, GDPR compliance, transparency reports
Battery drain on mobile/desktop	Medium	Medium	Optimize sampling rates, adaptive monitoring, user controls, battery impact testing in QA
Low user engagement after initial enthusiasm	High	Medium	Habit-building features, streak mechanics, meaningful insights, first-week guided onboarding
Cross-platform complexity leading to delays	High	High	Shared core logic, prioritize web + desktop for MVP, staged rollout, realistic timeline (see revised timeline)
Competition from established tools	Medium	Medium	Focus on integration and insights (see Competitive Analysis); our unique combination is the differentiator
Desktop app security concerns / antivirus false positives	Medium	High	EV code signing, antivirus vendor pre-submission, in-app help for false positive resolution, transparent permission explanations
Inaccurate activity classification undermining trust	High	Medium	Three-layer classification system, user overrides, confidence indicators, "Neutral" default for unknowns
Notification fatigue causing uninstalls	Medium	Medium	Unified notification manager, platform deduplication, granular user controls, quiet hours
Scope creep in MVP	High	High	Strict milestone gates; each milestone must be complete + tested before starting next; cut mobile from MVP if behind schedule
Data breach / security incident	Critical	Low	Encryption at rest + in transit, penetration testing, security audit, incident response plan, no raw keystrokes stored

Success Criteria for MVP Launch

- ✔ User can complete full onboarding flow in < 5 minutes
- ✔ User can start a focus session from 3+ platforms
- ✔ Desktop app accurately tracks activity with < 5% error rate

4. ☒ Dashboard displays real-time session progress
 5. ☒ Weekly analytics email delivers personalized insights
 6. ☒ 70% of beta users complete at least 3 sessions per week
 7. ☒ Average productivity score calculation matches user perception in qualitative feedback
 8. ☒ No critical bugs reported in activity tracking or data loss
 9. ☒ User data export works flawlessly (GDPR requirement)
 10. ☒ Positive feedback from 60%+ of beta testers on core value proposition
 11. ☒ **NEW:** All unit + integration tests passing with > 80% code coverage on backend
 12. ☒ **NEW:** Zero critical/high severity findings in security audit
 13. ☒ **NEW:** Accessibility audit passes WCAG 2.1 AA for web dashboard
 14. ☒ **NEW:** Desktop app passes antivirus scanning on Windows Defender, Avast, and Malwarebytes without false positives
 15. ☒ **NEW:** Offline → online sync completes without data loss in 95%+ of test scenarios
-

Timeline (Revised)

The original 19-week timeline was overly aggressive for the scope. This revised timeline adds buffer for testing, security, and the reality of cross-platform development.

Phase	Weeks	Duration	Deliverables
Phase 1: Foundation	1–4	4 weeks	Core web app, authentication, session engine, basic dashboard, database setup
Phase 2: Desktop & Browser	5–8	4 weeks	Desktop app (Windows) with activity tracking, Chrome browser extension, basic sync
Phase 3: Mobile & Breaks	9–12	4 weeks	Android app, break system, notification manager, cross-platform sync
Phase 4: Analytics	13–16	4 weeks	Classification engine, analytics dashboard, goal system, insights generation
Phase 5: Polish & Security	17–20	4 weeks	Onboarding flow, accessibility audit, security audit, performance optimization, antivirus testing
Phase 6: Testing & Beta	21–23	3 weeks	Full QA cycle, E2E testing, bug fixes, private beta launch (50-100 users)
Phase 7: Iteration	24–26	3 weeks	Beta feedback incorporation, final polish, documentation
Phase 8: Launch	27	1 week	Public launch, monitoring, incident response readiness

Total Revised MVP Development Time: ~6-7 months

Scope Reduction Contingency

If development falls behind schedule, cut in this order (least impact first):

- 1. **Cut mobile app** → Focus on web + desktop + browser extension (3 platforms is sufficient for MVP)
- 2. **Cut Firefox extension** → Ship Chrome-only for MVP, add Firefox post-launch
- 3. **Simplify analytics** → Rule-based classification only (no ML), basic charts (no heatmaps)
- 4. **Cut weekly email** → Dashboard-only analytics for MVP; add email reports post-launch

This PRD provides a comprehensive blueprint for building FlowShield. The key differentiator remains the "passive tracking + actionable insights" combination across multiple platforms, delivered through a privacy-first architecture. All new sections (Classification System, Notification Strategy, Offline Sync, Security Model, Testing Strategy, API Spec, User Personas, Competitive Analysis, Accessibility, Localization, Product Analytics, Account Management, and Data Retention) fill the gaps identified in the initial review.